Linear Functions & Inequalities

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Student-Produced Response Questions (SPRs): Foundational Fluency (1–7)

1 [Foundational] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

The graph of the linear function $y = f(x)$, which models the daily balance index at the municipal library, is shown in the xy-plane. Based on the two labeled points on the line, what is the value of $f(6)$?

Answer: -14. Reading the two marked points from the graph, the line passes through $(0, 4)$ and $(3, -5)$. The slope is $m = \frac{-5-4}{3-0} = -3$, so $f(x) = -3x + 4$. Evaluating at $x = 6$ gives $f(6) = -3(6) + 4 = -14$.

2 [Foundational] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

The graph of the linear function $y = f(x)$, which models the daily balance index at the coastal aquarium, is shown in the xy-plane. Based on the two labeled points on the line, what is the value of $f(6)$?

Answer: 19. Reading the two marked points from the graph, the line passes through $(0, 7)$ and $(3, 13)$. The slope is $m = \frac{13-7}{3-0} = 2$, so $f(x) = 2x + 7$. Evaluating at $x = 6$ gives $f(6) = 2(6) + 7 = 19$.

3 [Foundational] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

The graph of the linear function $y = f(x)$, which models the daily balance index at the urban bike-share station, is shown in the xy-plane. Based on the two labeled points on the line, what is the value of $f(6)$?

Answer: -27. Reading the two marked points from the graph, the line passes through $(0, 3)$ and $(3, -12)$. The slope is $m = \frac{-12-3}{3-0} = -5$, so $f(x) = -5x + 3$. Evaluating at $x = 6$ gives $f(6) = -5(6) + 3 = -27$.

4 [Foundational] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

The graph of the linear function $y = f(x)$, which models the daily balance index at the agricultural cooperative, is shown in the xy-plane. Based on the two labeled points on the line, what is the value of $f(6)$?

Answer: 30. Reading the two marked points from the graph, the line passes through $(0, 6)$ and $(3, 18)$. The slope is $m = \frac{18-6}{3-0} = 4$, so $f(x) = 4x + 6$. Evaluating at $x = 6$ gives $f(6) = 4(6) + 6 = 30$.

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5 [Foundational] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

The graph of the linear function $y = f(x)$, which models the daily balance index at the recycling center, is shown in the xy-plane. Based on the two labeled points on the line, what is the value of $f(6)$?

Answer: -9. Reading the two marked points from the graph, the line passes through $(0, 9)$ and $(3, 0)$. The slope is $m = \frac{0-9}{3-0} = -3$, so $f(x) = -3x + 9$. Evaluating at $x = 6$ gives $f(6) = -3(6) + 9 = -9$.

6 [Foundational] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

The graph of the linear function $y = f(x)$, which models the daily balance index at the neighborhood bookstore, is shown in the xy-plane. Based on the two labeled points on the line, what is the value of $f(6)$?

Answer: 17. Reading the two marked points from the graph, the line passes through $(0, 5)$ and $(3, 11)$. The slope is $m = \frac{11-5}{3-0} = 2$, so $f(x) = 2x + 5$. Evaluating at $x = 6$ gives $f(6) = 2(6) + 5 = 17$.

7 [Foundational] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

The graph of the linear function $y = f(x)$, which models the daily balance index at the culinary institute, is shown in the xy-plane. Based on the two labeled points on the line, what is the value of $f(6)$?

Answer: -22. Reading the two marked points from the graph, the line passes through $(0, 8)$ and $(3, -7)$. The slope is $m = \frac{-7-8}{3-0} = -5$, so $f(x) = -5x + 8$. Evaluating at $x = 6$ gives $f(6) = -5(6) + 8 = -22$.

Part I — Student-Produced Response Questions (SPRs): Intermediate Application (8–14)

8 [Intermediate] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

In a model for the maple syrup farm, what is the slope of the line in the xy-plane that passes through the points $(-3, 5)$ and $(1, -3)$?

Answer: -2. Slope $m = \frac{-3-5}{1-(-3)} = \frac{-8}{4} = -2$.

9 [Intermediate] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

In a model for the marine research vessel, in a calculation for the local farmers market (case 20), if $f(x) = 7x + 6$, what is $f(3)$?

Answer: 27. $f(3) = 7x + 6|_{x=3} = 7(3) + 6 = 27$

10 [Intermediate] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

In a model for the campus coffee roastery, in a calculation for the performing arts academy (case 24), if $f(x) = -7x - 4$, what is $f(6)$?

Answer: -46. $f(6) = -7x - 4|_{x=6} = -7(6) - 4 = -46$

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11 [Intermediate] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

In a model for the meteorological station, in a calculation for the coastal aquarium (case 28), if $f(x) = 7x - 4$, what is $f(1)$?

Answer: 3. $f(1) = 7x - 4|_{x=1} = 7(1) - 4 = 3$

12 [Intermediate] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

In a model for the urban bike-share station, what is the slope of the line passing through $(2, 5)$ and $(6, 17)$?

Answer: 3. $m = \frac{17-5}{6-2} = \frac{12}{4} = 3$.

13 [Intermediate] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

In a model for the art gallery, in the coordinate plane, the graph of $y = H(y)$ is translated 8 units right and 9 units up to form $y = K(y)$, mapping $E(12, 61)$ to F. What is the sum of coordinates of F on $y = K(y)$?

Answer: 90. The image point is $F(20, 70)$, whose coordinates sum to 90.

14 [Intermediate] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

In a model for the sports complex, a line is parallel to the line $y = 2x + 3$ and passes through the point $(7, -4)$. What is its y-intercept?

Answer: -18. Parallel to lines have slope 2. $-4 = 2(7) + b \Rightarrow b = -18$.

Part I — Student-Produced Response Questions (SPRs): Challenging Synthesis (15–20)

15 [Challenging] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

In a model for the regional airport shuttle, let the functions U and V be defined by $U(v) = 23v - 15$ and $V(v) = v^2 + 7$. What is the composite value $U(V(7))$?

Answer: 1273. First compute $V(7) = 7^2 + 7 = 56$. Then $U(56) = 23(56) - 15 = 1273$.

16 [Challenging] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

The botanical conservatory orders adult tickets at USD 4 per unit plus a flat shipping fee of USD 15. If the total order cost cannot exceed USD 38, what is the maximum whole number of adult tickets that can be ordered?

Answer: 5. Let n be the number of adult tickets. The inequality is $4n + 15 \le 38 \Rightarrow 4n \le 23 \Rightarrow n \le 5.75$. Since n must be a whole number, the maximum value is 5.

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17 [Challenging] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

In the xy-plane map of the municipal library, line k is perpendicular to the line $y = 3x + 3$ and passes through the point $(6, 6)$. What is the y-intercept of line k?

Answer: 8. The slope of the given line is 3, so the perpendicular line k has slope $m_k = -\frac{1}{3}$. Using point-slope form $y - 6 = -\frac{1}{3}(x - 6)$ and setting $x = 0$ gives $y = 6 + \frac{6}{3} = 8$.

18 [Challenging] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

A water storage tank at the community theater initially holds 130 gallons and is filled at a constant rate of 10 gallons per minute. How many gallons of water are in the tank after 6 minutes?

Answer: 190. Using the linear function $W(t) = 130 + 10t$, substituting $t = 6$ gives $W(6) = 130 + 10(6) = 190$ gallons.

19 [Challenging] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

The regional bakery orders standard passes at USD 7 per unit plus a flat shipping fee of USD 18. If the total order cost cannot exceed USD 80, what is the maximum whole number of standard passes that can be ordered?

Answer: 8. Let n be the number of standard passes. The inequality is $7n + 18 \le 80 \Rightarrow 7n \le 62 \Rightarrow n \le 8.857142857142858$. Since n must be a whole number, the maximum value is 8.

20 [Challenging] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

In the xy-plane map of the coastal aquarium, line k is perpendicular to the line $y = 6x + 6$ and passes through the point $(12, 9)$. What is the y-intercept of line k?

Answer: 11. The slope of the given line is 6, so the perpendicular line k has slope $m_k = -\frac{1}{6}$. Using point-slope form $y - 9 = -\frac{1}{6}(x - 12)$ and setting $x = 0$ gives $y = 9 + \frac{12}{6} = 11$.

Part II — Multiple-Choice Questions (MCQs): Foundational Fluency (21–28)

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21 [Foundational] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

The graph of the linear function $y = f(x)$, which models the daily balance index at the botanical conservatory, is shown in the xy-plane. Based on the two labeled points on the line, what is the value of $f(5)$?

a. 13 ✓

b. 15

c. 10

d. 16

Reading the two marked points from the graph, the line passes through $(0, 3)$ and $(2, 7)$. The slope is $m = \frac{7-3}{2-0} = 2$, so $f(x) = 2x + 3$. Evaluating at $x = 5$ gives $f(5) = 2(5) + 3 = 13$.

22 [Foundational] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

The graph of the linear function $y = f(x)$, which models the daily balance index at the community theater, is shown in the xy-plane. Based on the two labeled points on the line, what is the value of $f(7)$?

a. 28

b. 26

c. 33 ✓

d. 37

Reading the two marked points from the graph, the line passes through $(0, 5)$ and $(4, 21)$. The slope is $m = \frac{21-5}{4-0} = 4$, so $f(x) = 4x + 5$. Evaluating at $x = 7$ gives $f(7) = 4(7) + 5 = 33$.

23 [Foundational] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

The graph of the linear function $y = f(x)$, which models the daily balance index at the regional bakery, is shown in the xy-plane. Based on the two labeled points on the line, what is the value of $f(5)$?

a. 2

b. -19 ✓

c. -24

d. -25

Reading the two marked points from the graph, the line passes through $(0, 6)$ and $(2, -4)$. The slope is $m = \frac{-4-6}{2-0} = -5$, so $f(x) = -5x + 6$. Evaluating at $x = 5$ gives $f(5) = -5(5) + 6 = -19$.

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24 [Foundational] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

The graph of the linear function $y = f(x)$, which models the daily balance index at the mountain observatory, is shown in the xy-plane. Based on the two labeled points on the line, what is the value of $f(7)$?

a. -16

b. -21

c. 4

d. -13 ✓

Reading the two marked points from the graph, the line passes through $(0, 8)$ and $(4, -4)$. The slope is $m = \frac{-4-8}{4-0} = -3$, so $f(x) = -3x + 8$. Evaluating at $x = 7$ gives $f(7) = -3(7) + 8 = -13$.

25 [Foundational] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

The graph of the linear function $y = f(x)$, which models the daily balance index at the science museum, is shown in the xy-plane. Based on the two labeled points on the line, what is the value of $f(5)$?

a. 20

b. 26

c. 29 ✓

d. 33

Reading the two marked points from the graph, the line passes through $(0, 9)$ and $(2, 17)$. The slope is $m = \frac{17-9}{2-0} = 4$, so $f(x) = 4x + 9$. Evaluating at $x = 5$ gives $f(5) = 4(5) + 9 = 29$.

26 [Foundational] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

The graph of the linear function $y = f(x)$, which models the daily balance index at the solar microgrid facility, is shown in the xy-plane. Based on the two labeled points on the line, what is the value of $f(7)$?

a. 18 ✓

b. 20

c. 14

d. 16

Reading the two marked points from the graph, the line passes through $(0, 4)$ and $(4, 12)$. The slope is $m = \frac{12-4}{4-0} = 2$, so $f(x) = 2x + 4$. Evaluating at $x = 7$ gives $f(7) = 2(7) + 4 = 18$.

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27 [Foundational] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

The graph of the linear function $y = f(x)$, which models the daily balance index at the ceramic studio, is shown in the xy-plane. Based on the two labeled points on the line, what is the value of $f(5)$?

a. -13

b. -15

c. 4

d. -10 ✓

Reading the two marked points from the graph, the line passes through $(0, 5)$ and $(2, -1)$. The slope is $m = \frac{-1-5}{2-0} = -3$, so $f(x) = -3x + 5$. Evaluating at $x = 5$ gives $f(5) = -3(5) + 5 = -10$.

28 [Foundational] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

The graph of the linear function $y = f(x)$, which models the daily balance index at the publishing house, is shown in the xy-plane. Based on the two labeled points on the line, what is the value of $f(7)$?

a. -6

b. -28 ✓

c. -33

d. -35

Reading the two marked points from the graph, the line passes through $(0, 7)$ and $(4, -13)$. The slope is $m = \frac{-13-7}{4-0} = -5$, so $f(x) = -5x + 7$. Evaluating at $x = 7$ gives $f(7) = -5(7) + 7 = -28$.

Part II — Multiple-Choice Questions (MCQs): Intermediate Application (29–37)

29 [Intermediate] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

In a model for the art gallery, in a calculation for the wildlife sanctuary (case 11), if $f(x) = 7x - 11$, what is $f(-9)$?

a. -67

b. -74 ✓

c. -83

d. -81

$f(-9) = 7x - 11|_{x=-9} = 7(-9) - 11 = -74$

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30 [Intermediate] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

In a model for the robotics club, which of the following lines is parallel to the line $y = 4x - 9$ in the xy-plane?

a. $y = 4x + 12$ ✓

b. $y = -4x - 9$

c. $y = -\frac{1}{4}x + 3$

d. $y = \frac{1}{4}x - 9$

Parallel lines have identical slopes. The line $y = 4x - 9$ has slope 4, so $y = 4x + 12$ is parallel to it.

31 [Intermediate] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

In a model for the culinary institute, what is the slope of any line in the xy-plane that is perpendicular to the line $y = -\frac{2}{5}x + 3$?

a. $\frac{5}{2}$ ✓

b. $-\frac{5}{2}$

c. $\frac{2}{5}$

d. $-\frac{2}{5}$

The slope of the given line is $-\frac{2}{5}$. Perpendicular lines have negative reciprocal slopes: $-\frac{1}{-2/5} = \frac{5}{2}$.

32 [Intermediate] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

In a model for the water treatment plant, a line has slope -8 and passes through the point $(3, -9)$. What is the y-intercept of the line?

a. 24

b. 9

c. 20

d. 15 ✓

$y = mx + b \Rightarrow -9 = -8(3) + b \Rightarrow b = 15$

33 [Intermediate] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

In a model for the sports complex, in the coordinate plane, the graph of $y = P(r)$ is translated 13 units right and 3 units up to form $y = Q(r)$, mapping $U(6, 67)$ to V. What is the sum of coordinates of V on $y = Q(r)$?

a. 76

b. 89 ✓

c. 92

d. 105

The image point is $V(19, 70)$, whose coordinates sum to 89.

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34 [Intermediate] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

In a model for the planetarium, a gym charges a monthly membership fee of USD 30 plus USD 2 for each fitness class attended. If a member attends c fitness classes in a month, which expression represents the member's total cost for that month?

a. $30 + 2c$ ✓

b. $30c + 2$

c. $2(30 + c)$

d. $\frac{30 + c}{2}$

The base monthly fee is 30 and the variable fee is $2c$, giving $30 + 2c$.

35 [Intermediate] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

In a model for the regional airport shuttle, the point $(4, 14)$ lies on the line $y = kx - 2$ in the xy-plane, where k is a constant. What is the value of k?

a. 3

b. 4 ✓

c. 5

d. 6

Substitute $x = 4$ and $y = 14$: $14 = 4k - 2 \Rightarrow 4k = 16 \Rightarrow k = 4$.

36 [Intermediate] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

In a model for the wind energy cooperative, what is the slope of the horizontal line defined by $y = 7$ in the xy-plane?

a. 0 ✓

b. 1

c. 7

d. undefined

A horizontal line has zero vertical change ($\Delta y = 0$), so its slope is 0.

37 [Intermediate] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

In a model for the national park visitor center, a line has slope -7 and passes through the point $(-2, 12)$. What is the y-intercept of the line?

a. -9

b. -2 ✓

c. 1

d. -7

$y = mx + b \Rightarrow 12 = -7(-2) + b \Rightarrow b = -2$

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Part II — Multiple-Choice Questions (MCQs): Challenging Synthesis (38–45)

38 [Challenging] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

In a model for the maple syrup farm, a line is perpendicular to $y = -\frac{2}{3}x + 5$ and has a y-intercept of $(0, -4)$. What is the equation of this line?

a. $y = \frac{3}{2}x - 4$ ✓

b. $y = -\frac{3}{2}x - 4$

c. $y = \frac{2}{3}x - 4$

d. $y = -\frac{2}{3}x - 4$

The perpendicular slope is the negative reciprocal of $-2/3$, which is $3/2$. With y-intercept -4, the equation is $y = \frac{3}{2}x - 4$.

39 [Challenging] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

In a model for the planetarium, which point lies in the solution set of the system of inequalities? $\{y > 2x - 3; y \leq -x + 4$

a. $(2, 5)$

b. $(1, 2)$ ✓

c. $(3, 2)$

d. $(0, -4)$

Test $(1, 2)$: $2 > 2(1) - 3 = -1$ (True) and $2 \leq -1 + 4 = 3$ (True).

40 [Challenging] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

In a model for the wind energy cooperative, line k passes through $(-3, 7)$ and $(5, -9)$. What is the x-intercept of line k?

a. $(0.5, 0)$ ✓

b. $(1, 0)$

c. $(0, 1)$

d. $(2, 0)$

Slope $m = \frac{-9-7}{5-(-3)} = \frac{-16}{8} = -2$. Equation: $y - 7 = -2(x + 3) \Rightarrow y = -2x + 1$. For x-intercept, set $y = 0$: $0 = -2x + 1 \Rightarrow 2x = 1 \Rightarrow x = 0.5$.

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41 [Challenging] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

In a model for the marine research vessel, which of the following lines is perpendicular to the line $5x + 2y = 10$?

a. $2x - 5y = 15$ ✓

b. $5x - 2y = 10$

c. $2x + 5y = 20$

d. $-5x - 2y = 8$

Slope of $5x + 2y = 10$ is $-5/2$. The perpendicular slope is $2/5$. Equation $2x - 5y = 15 \Rightarrow 5y = 2x - 15 \Rightarrow y = \frac{2}{5}x - 3$, which has slope $2/5$.

42 [Challenging] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

The water treatment plant orders fiction novels at USD 4 per unit plus a flat shipping fee of USD 36. If the total order cost cannot exceed USD 99, what is the maximum whole number of fiction novels that can be ordered?

a. 16

b. 14

c. 17

d. 15 ✓

Let n be the number of fiction novels. The inequality is $4n + 36 \le 99 \Rightarrow 4n \le 63 \Rightarrow n \le 15.75$. Since n must be a whole number, the maximum value is 15.

43 [Challenging] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

In the xy-plane map of the orchestra hall, line k is perpendicular to the line $y = 4x + 3$ and passes through the point $(8, 27)$. What is the y-intercept of line k?

a. 33

b. 27

c. 29 ✓

d. 25

The slope of the given line is 4, so the perpendicular line k has slope $m_k = -\frac{1}{4}$. Using point-slope form $y - 27 = -\frac{1}{4}(x - 8)$ and setting $x = 0$ gives $y = 27 + \frac{8}{4} = 29$.

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44 [Challenging] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

A water storage tank at the sports complex initially holds 235 gallons and is filled at a constant rate of 13 gallons per minute. How many gallons of water are in the tank after 9 minutes?

a. 365

b. 352 ✓

c. 244

d. 339

Using the linear function $W(t) = 235 + 13t$, substituting $t = 9$ gives $W(9) = 235 + 13(9) = 352$ gallons.

45 [Challenging] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

The charter ferry service orders hiking packs at USD 7 per unit plus a flat shipping fee of USD 39. If the total order cost cannot exceed USD 94, what is the maximum whole number of hiking packs that can be ordered?

a. 7 ✓

b. 8

c. 6

d. 9

Let n be the number of hiking packs. The inequality is $7n + 39 \le 94 \Rightarrow 7n \le 55 \Rightarrow n \le 7.857142857142857$. Since n must be a whole number, the maximum value is 7.

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